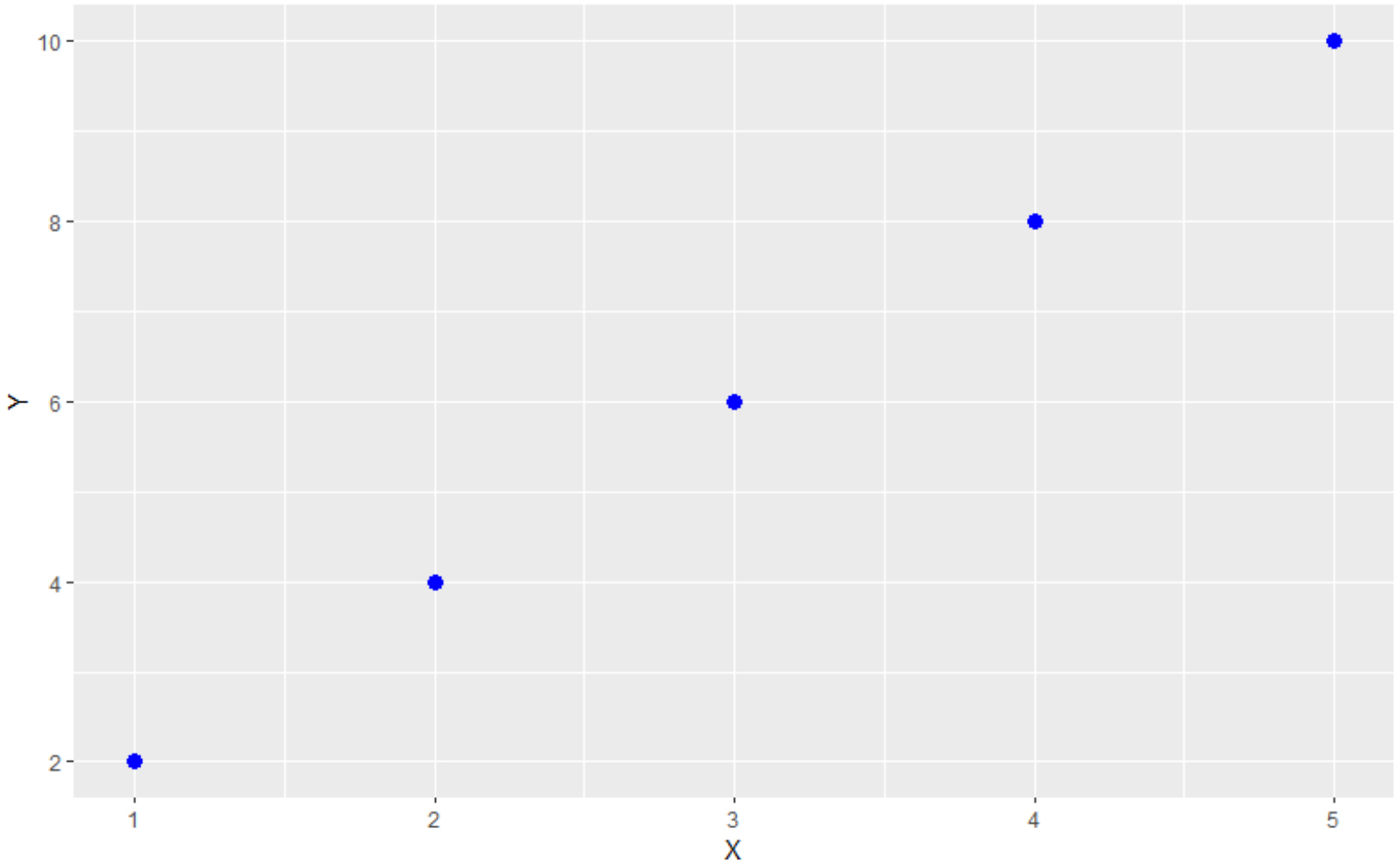
Covariation and Correlation

October 24, 2022

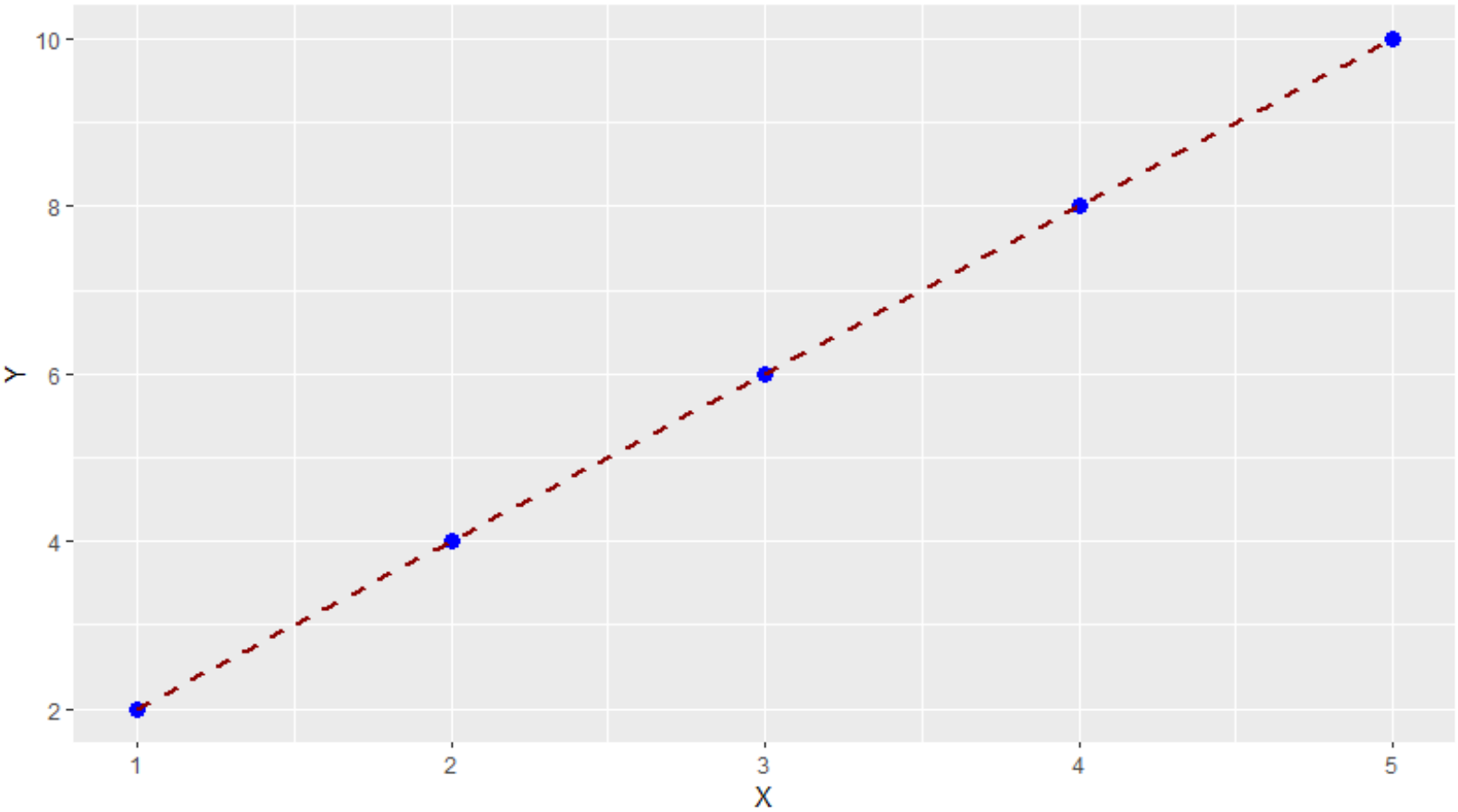
Bivariate Hypothesis Testing

	Independent Variable		
Dependent Variable		*Categorical*	*Continuous*
	Categorical	Crosstab/Chi-square	Logit/Probit
	Continuous	Difference of Means/T-test; Bivariate Regression	Correlation Coefficient; Bivariate Regression

Scatterplot



Scatterplot



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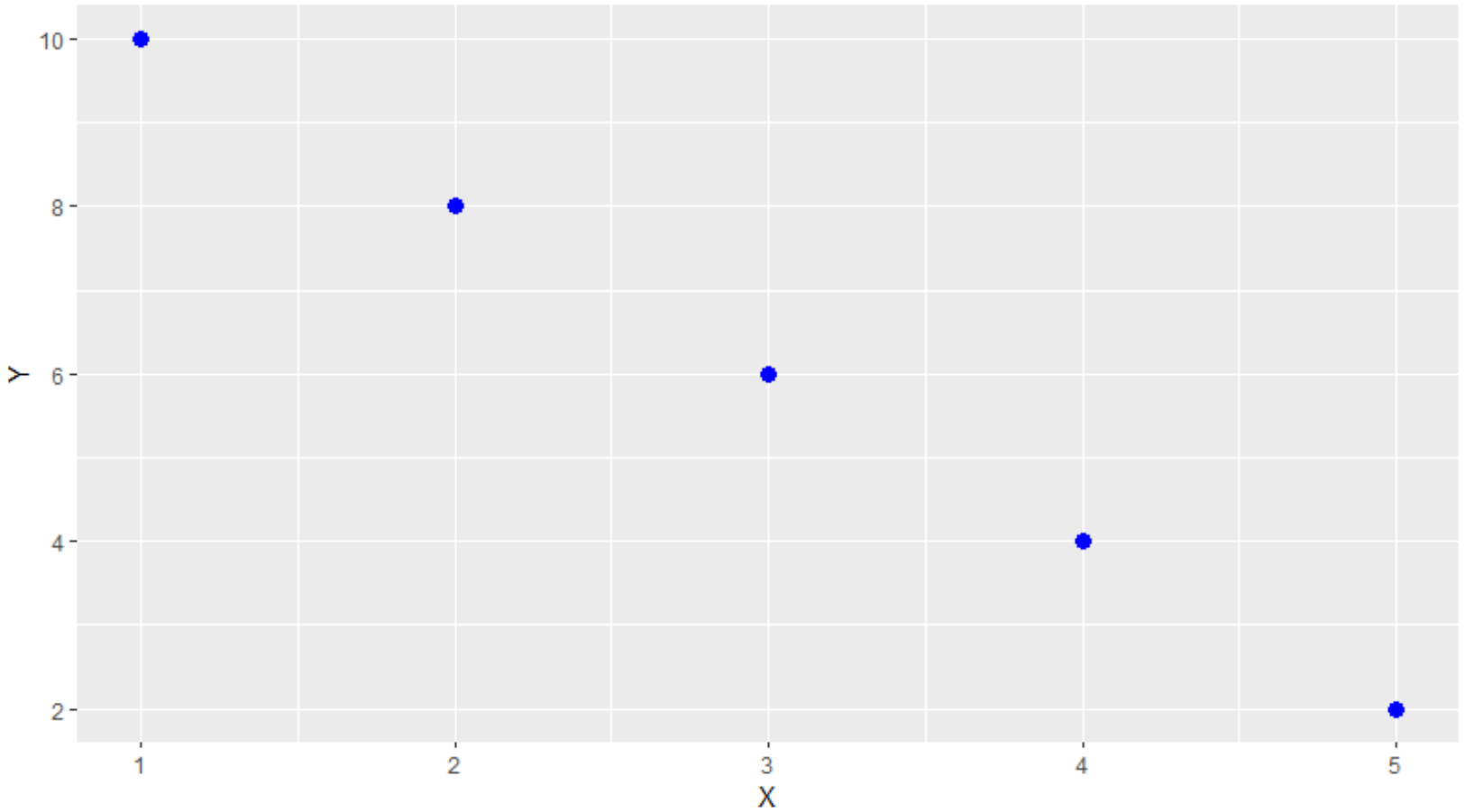

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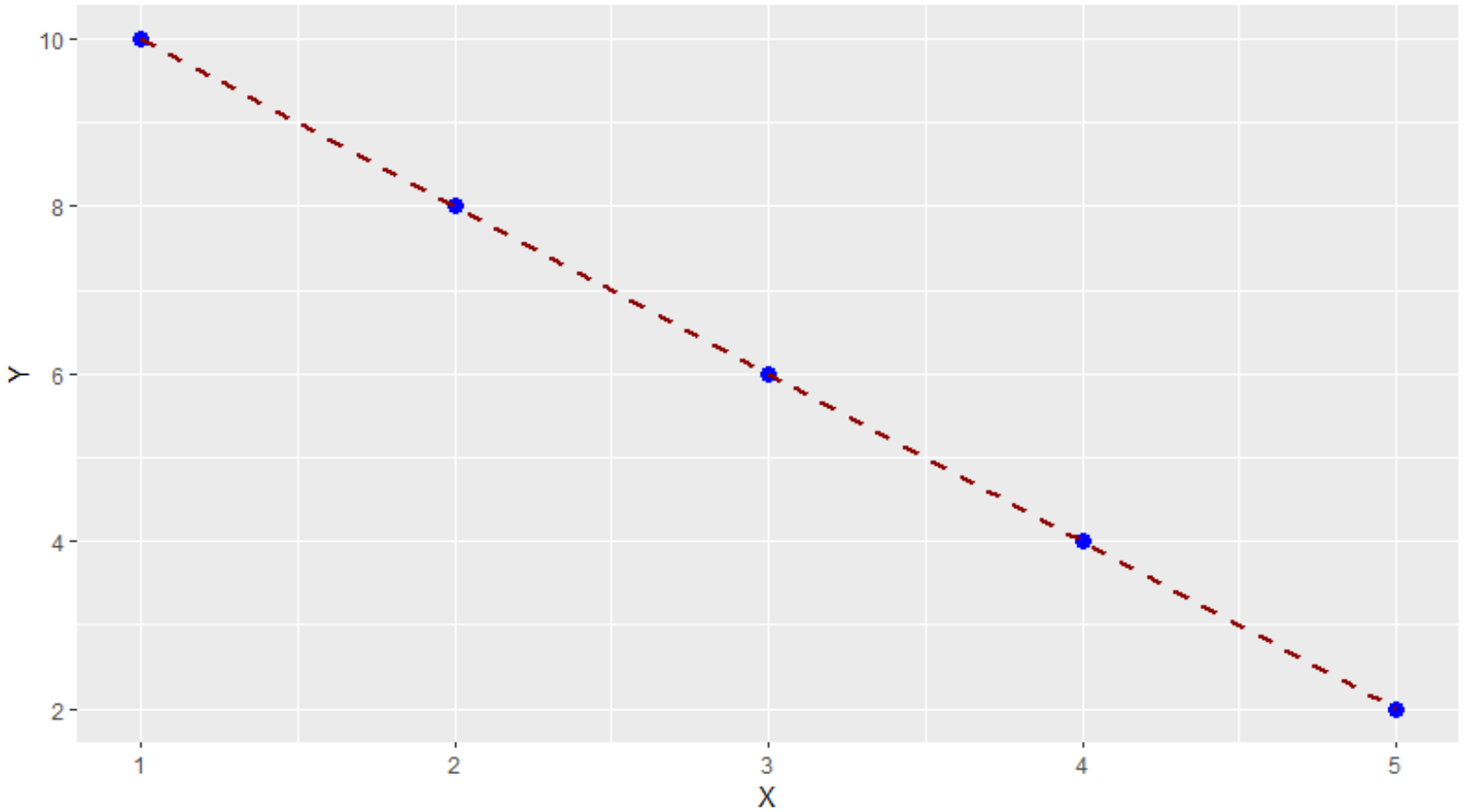

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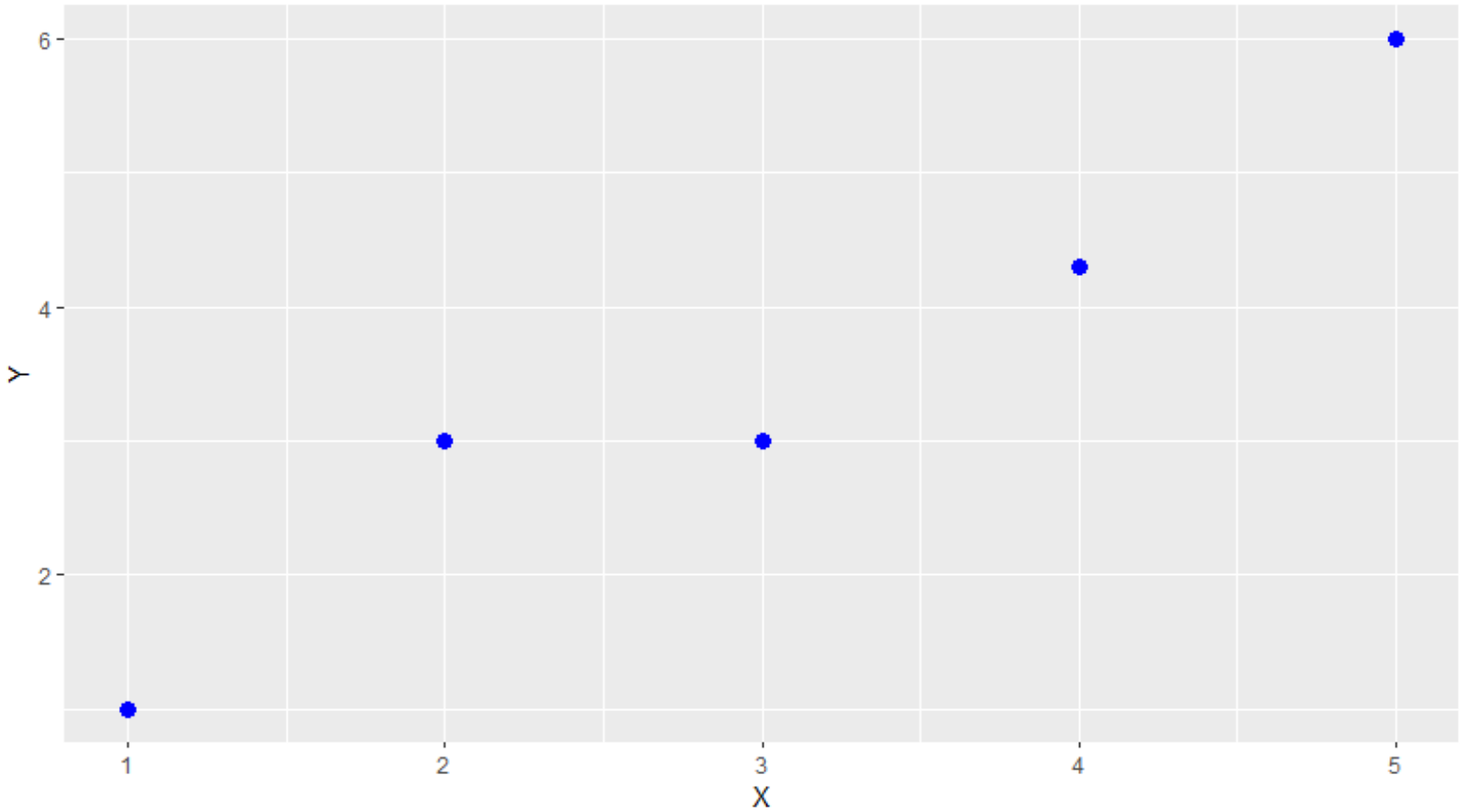

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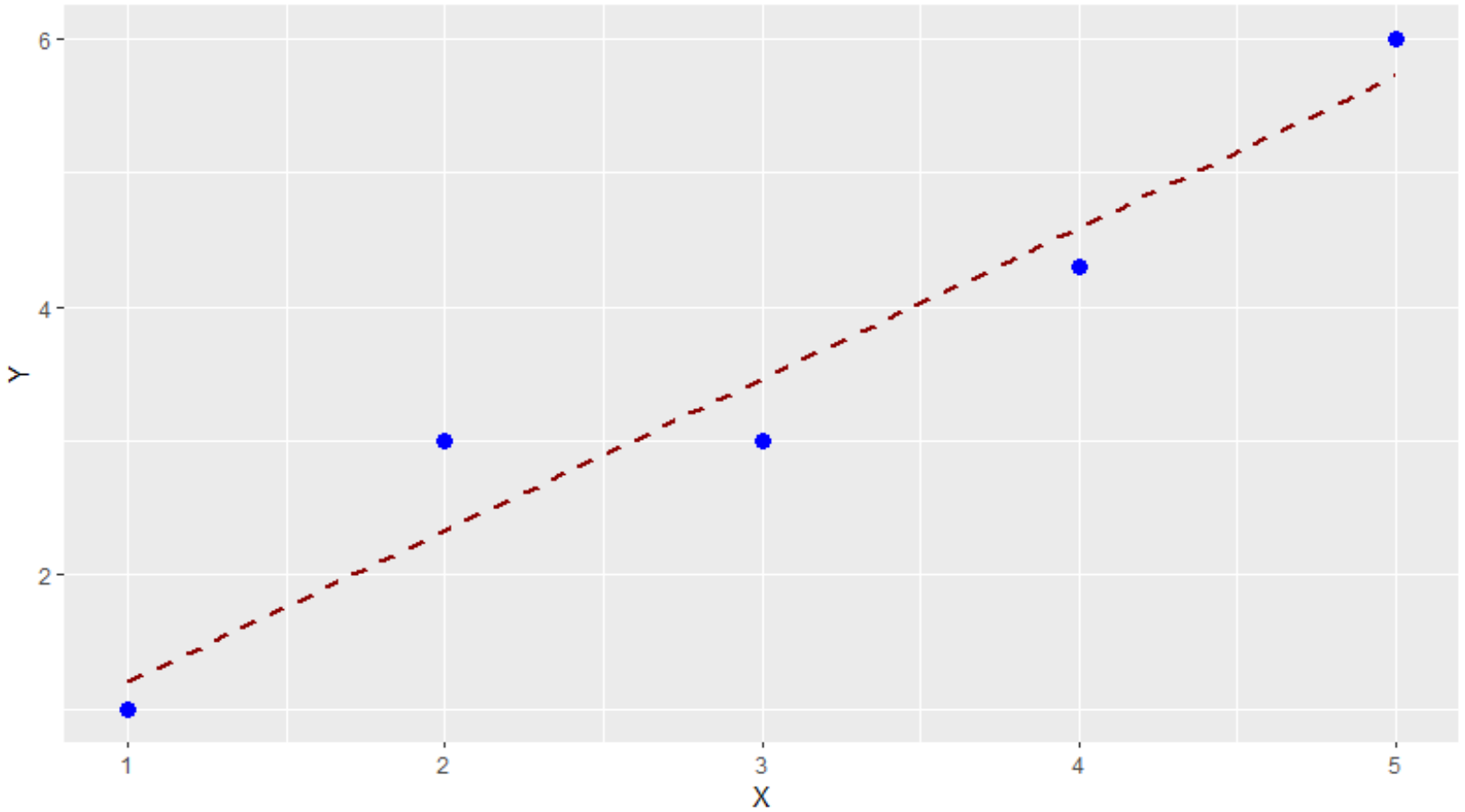

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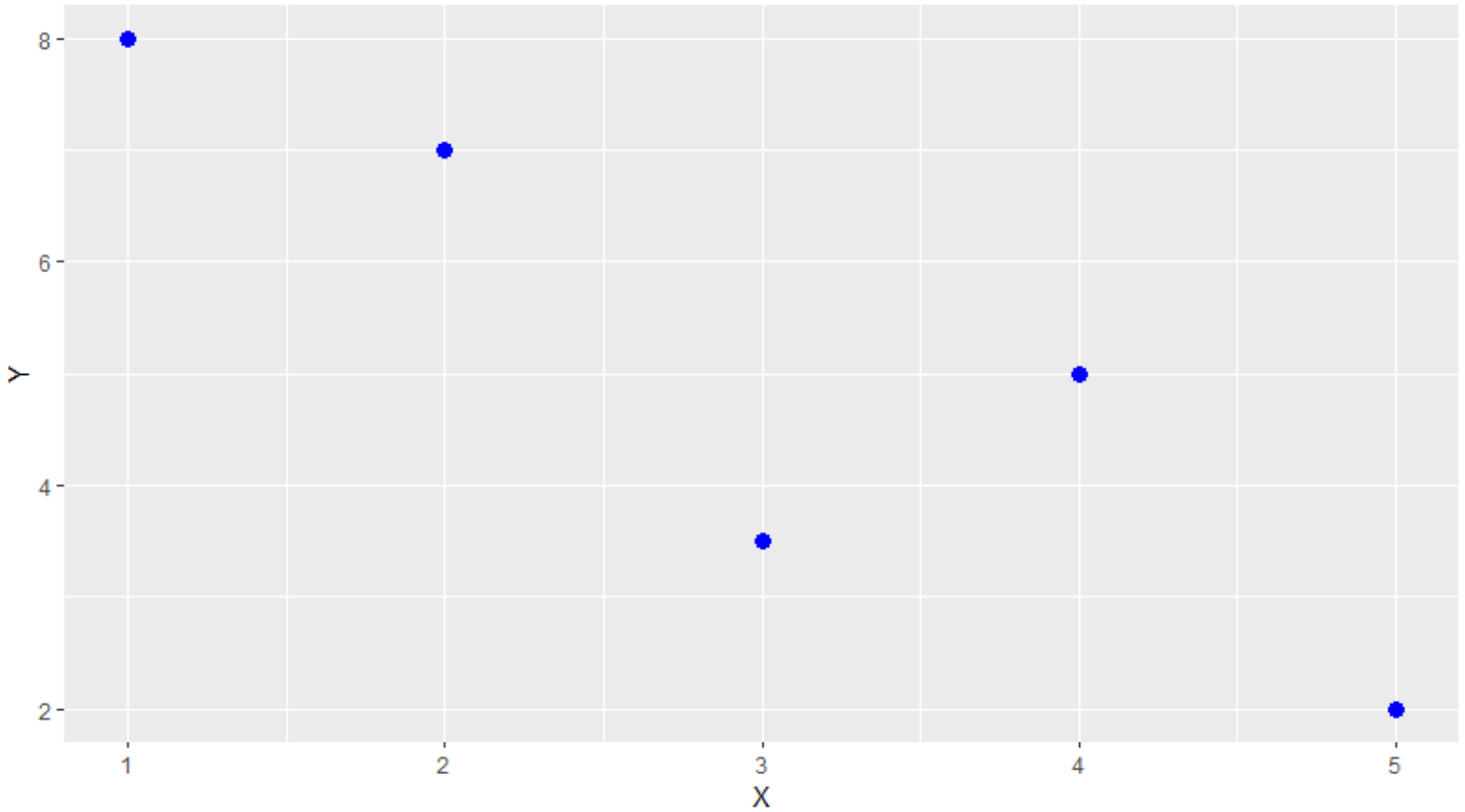

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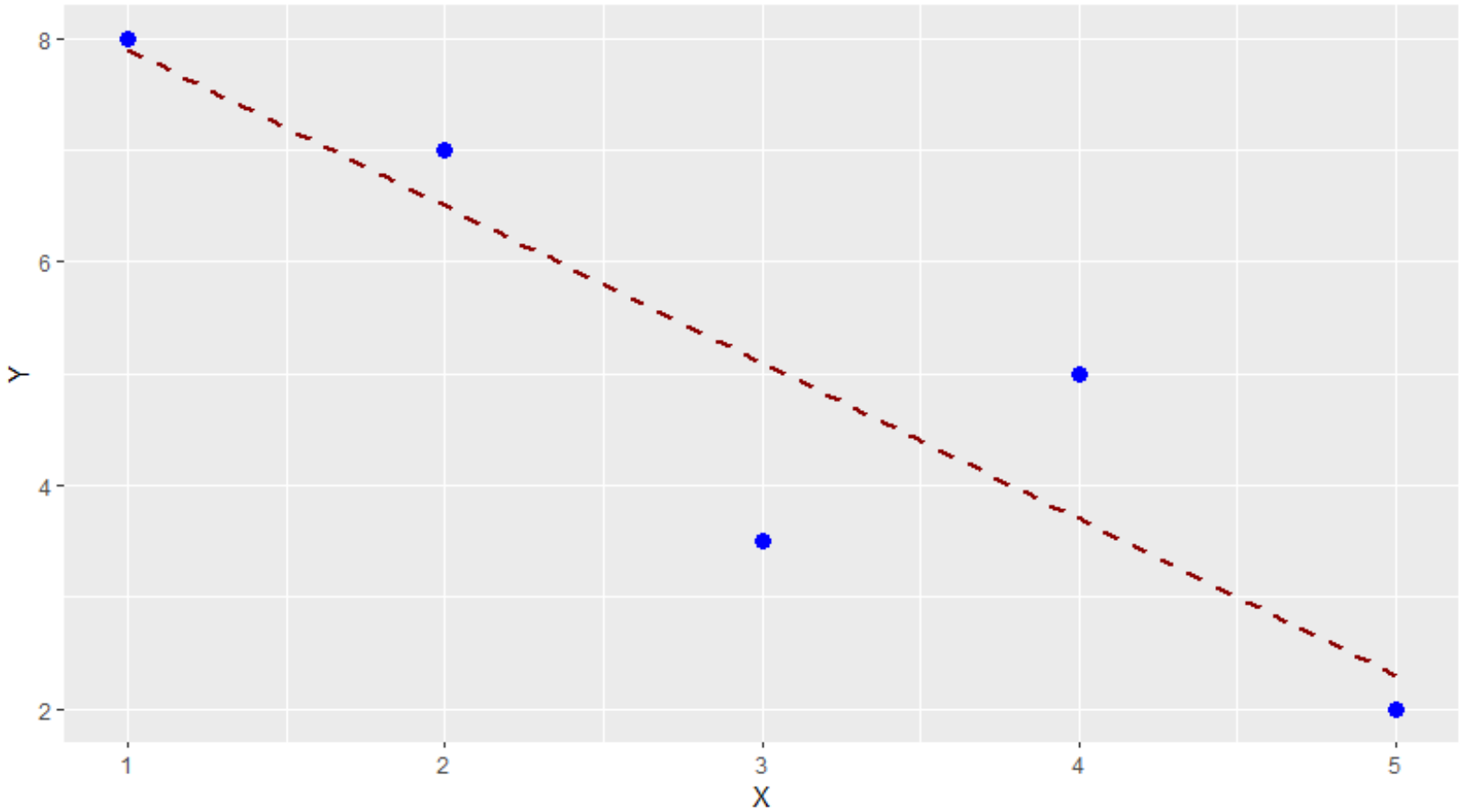


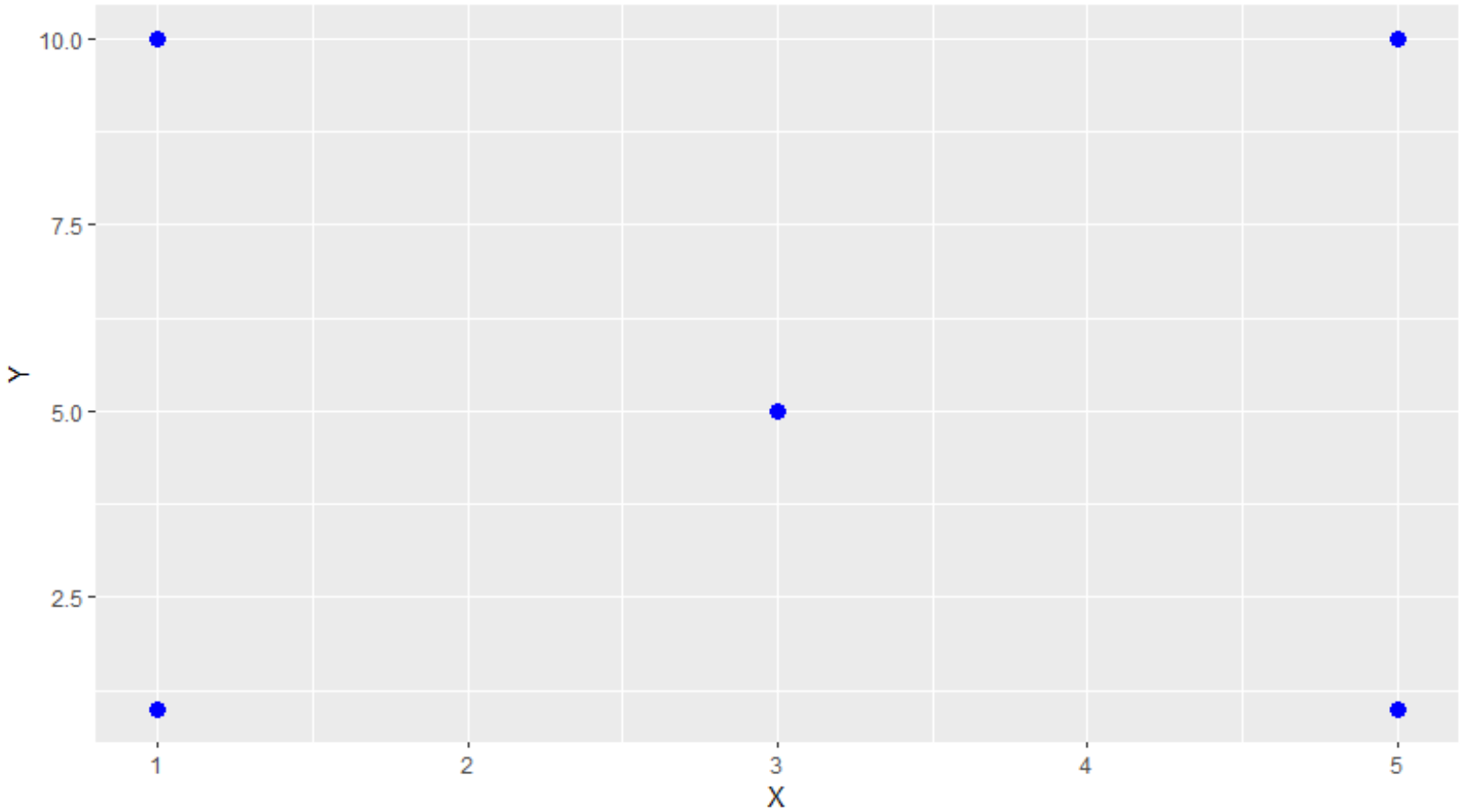
Covariance

An unstandardized statistical measure summarizing the general pattern of association (or the lack thereof) between two continuous variables.

- Positive number means positive relationship
- Negative number means negative relationship

Covariance $= 0$?

Covariance = 0?

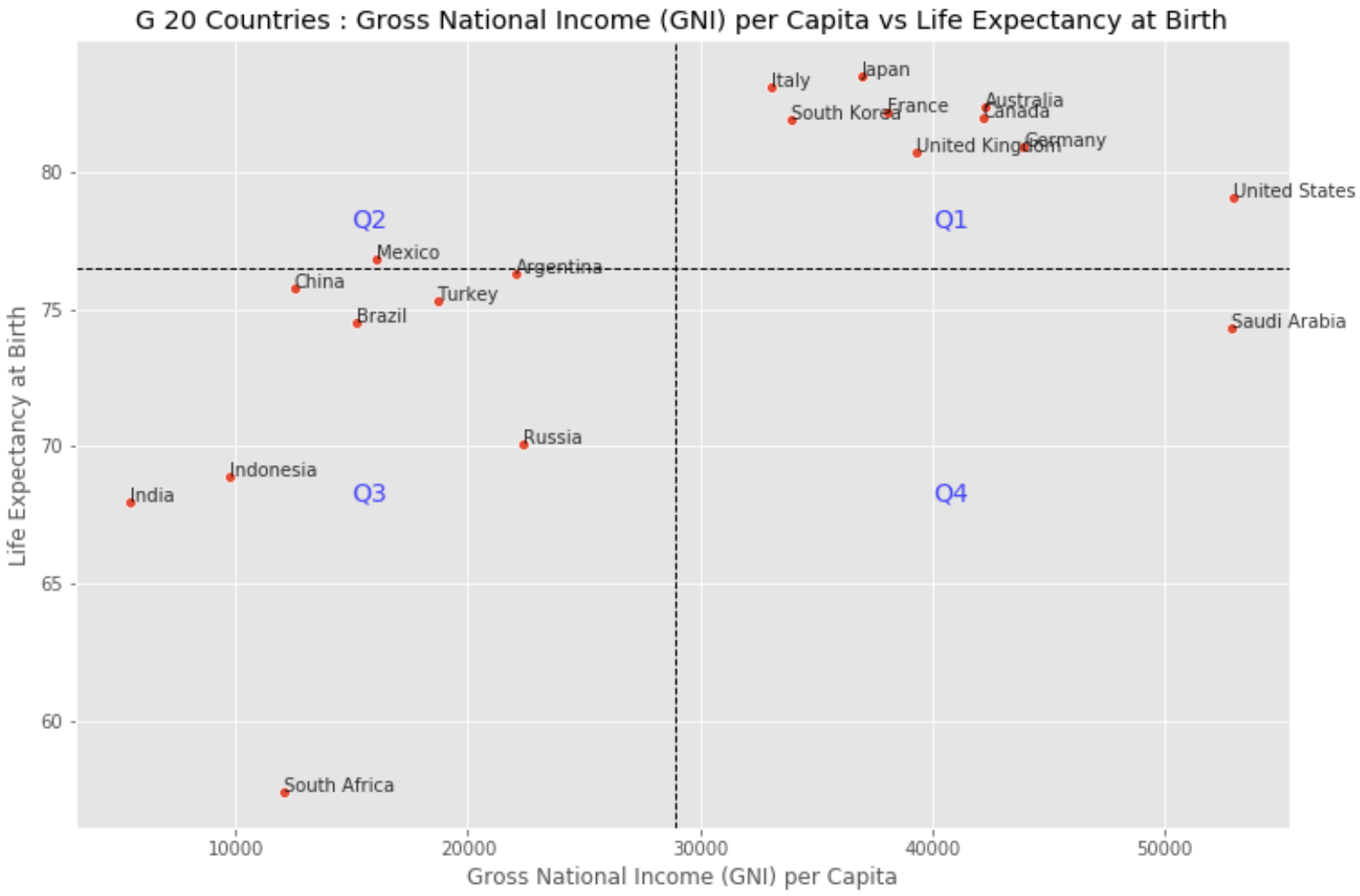


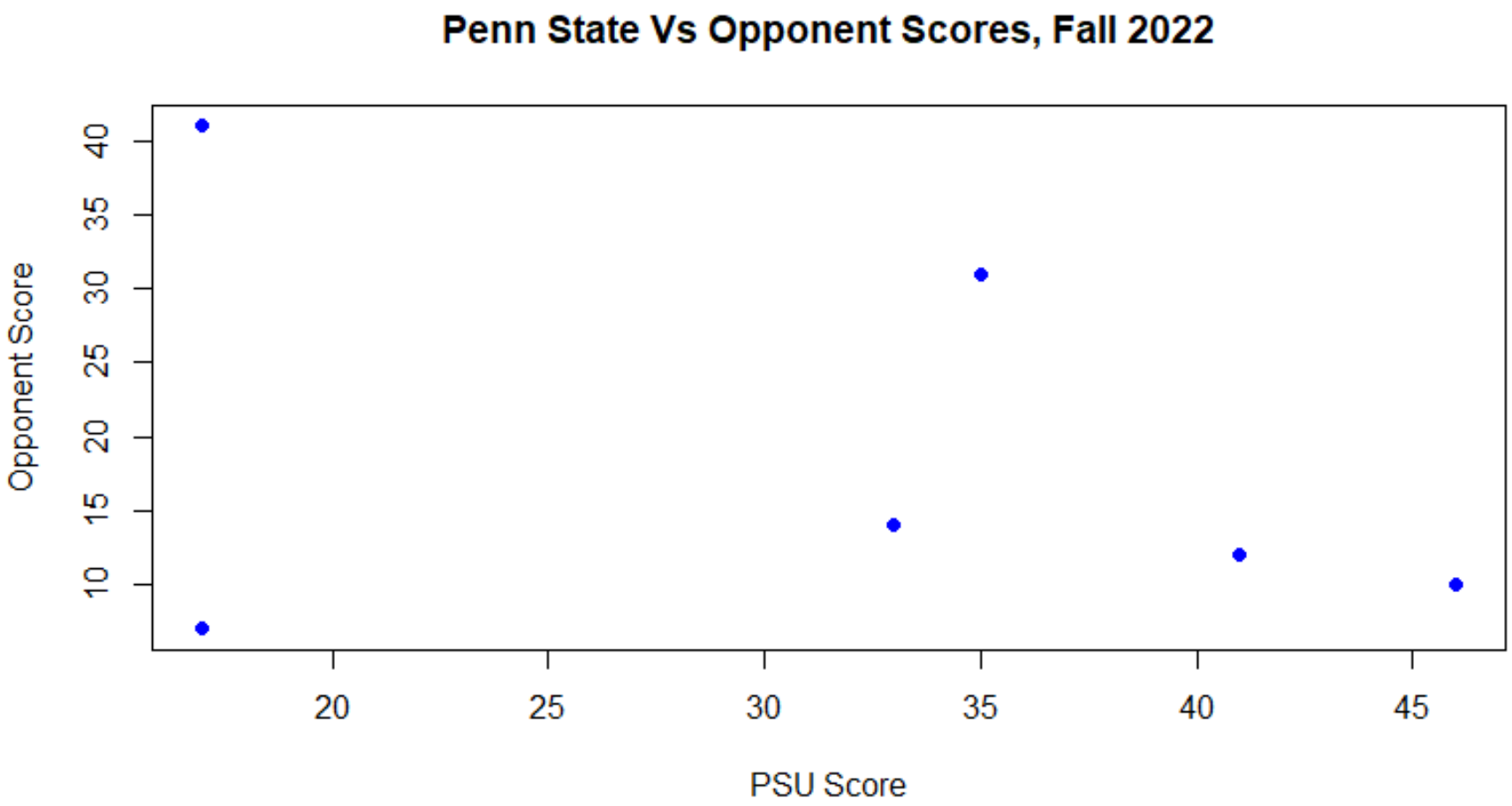
Covariance

$$cov_{XY} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{n-1} \tag{1}$$

- X_i = Value for X for each case i
- Y_i = Value for Y for each case i
- $\bar{X}$ = Mean of all X values
- $\bar{Y}$ = Mean of all Y values
- n = Sample size

$$cov_{XY} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$




Penn State Vs Opponent Scores, Fall 2022
Opponent Score
10
15
20
25
30
35
40
20
25
30
35
40
45
PSU Score

Opponent	PSU Score	Opponent Score
Purdue	35	31
Ohio	46	10
Auburn	41	12
Central Michigan	33	14
Northwestern	17	7
Michigan	41	17

Opponent	PSU Score	Opponent Score	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
Purdue	35	31	3.5	12	42
Ohio	46	10	14.5	-9	-130.5
Auburn	41	12	9.5	-7	-66.5
Central Michigan	33	14	1.5	-5	-7.5
Northwestern	17	7	-14.5	-12	174
Michigan	17	41	-14.5	22	-319
Mean	31.5	19 (approx.)			Sum =

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Mean	31.5	19 (approx.)			Sum = **-307.5**

$$cov_{XY} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{n-1} = \frac{-307.5}{6-1} =$$

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Is this a *strong* correlation?

$$cov_{XY} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{n-1} = \frac{-307.5}{6-1} = \textbf{-61.5}$$

Is this a *strong* correlation?

Can we confidently say that this is different from 0 (e.g., no correlation)?

Pearson's R (a.k.a. Correlation Coefficient)

$$r = \frac{cov_{XY}}{\sqrt{var_X var_Y}} \tag{2}$$

- cov_{XY} = Covariance of X and Y
- var_X = Variance of X
- var_Y = Variance of Y

Pearson's R (a.k.a. Correlation Coefficient)

$$r = \frac{cov_{XY}}{\sqrt{var_X var_Y}} = \frac{-16.5}{\sqrt{141.7 \times 185.37}} =$$

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$$r = \frac{cov_{XY}}{\sqrt{var_X var_Y}} = \frac{-16.5}{\sqrt{141.7 \times 185.37}} = \mathbf{-0.37}$$

Thresholds (arbitrary):

- $r > 0.8$ = strong positive linear correlation
- $0.8 > r > 0.5$ = decent positive linear correlation
- $0.5 > r > 0.3$ = weak positive linear correlation
- (reverse for negative r)

Hypothesis Testing with r

Null: $r = 0$

Alternative: $r \neq 0$
We can calculate a t-statistic to determine if we can reject the null that there is no linear relationship between X and Y:

$$t_r = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \tag{3}$$

With $n - 2$ degrees of freedom.

TABLE 5.1: Part of Table B Displaying t-Scores. The scores have right-tail probabilities of 0.100, 0.050, 0.025, 0.010, 0.005, and 0.001.

	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
1	3.078	6.314	12.706	31.821	63.657	318.3
10	1.372	1.812	2.228	2.764	3.169	4.144
28	1.313	1.701	2.048	2.467	2.763	3.408
30	1.310	1.697	2.042	2.457	2.750	3.385
100	1.290	1.660	1.984	2.364	2.626	3.174
infinity	1.282	1.645	1.960	2.326	2.576	3.090

Hypothesis Test

Since my test statistic value (0.797) is smaller than my critical value, we fail to reject the null hypothesis.

Final Thoughts

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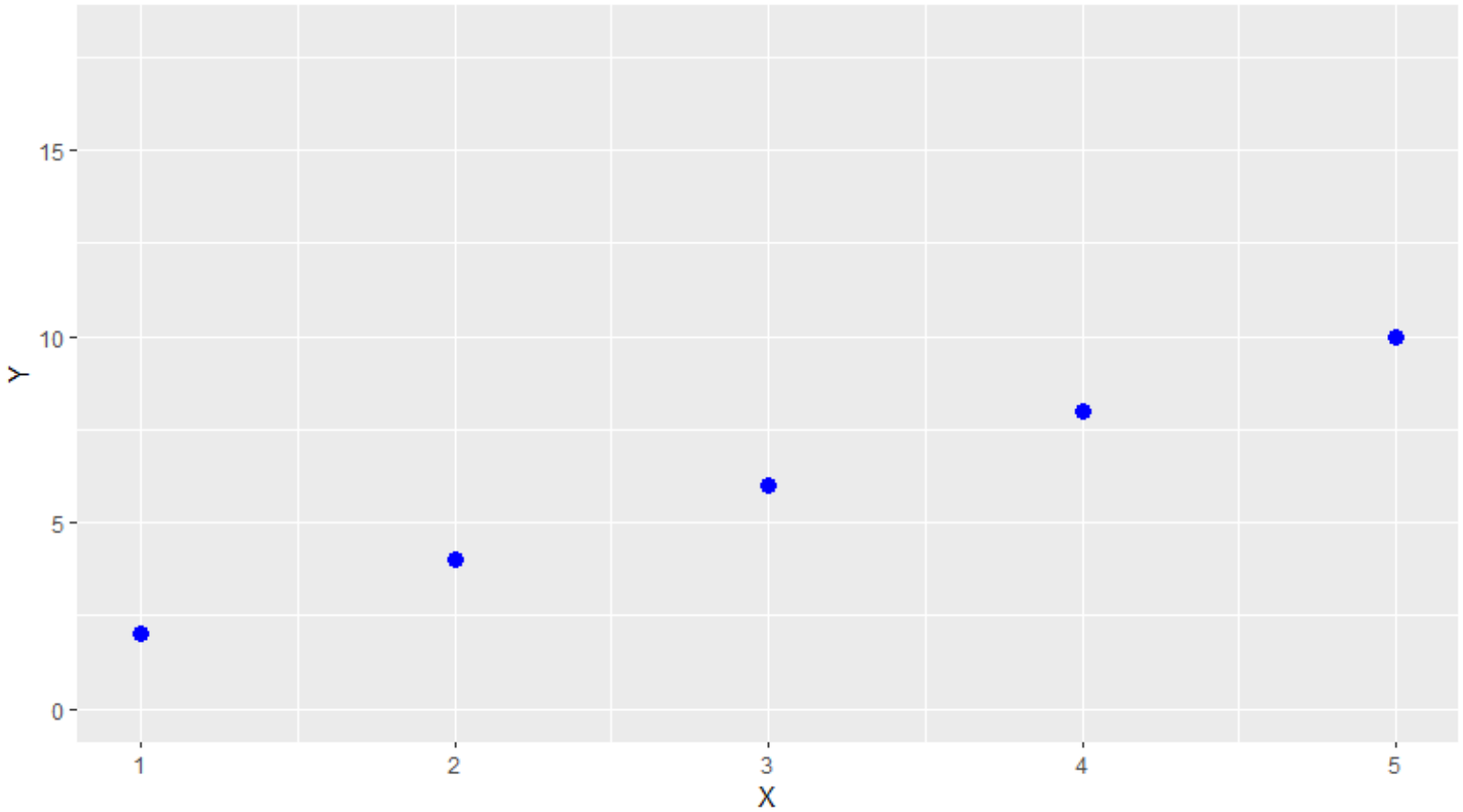
$r = 1$



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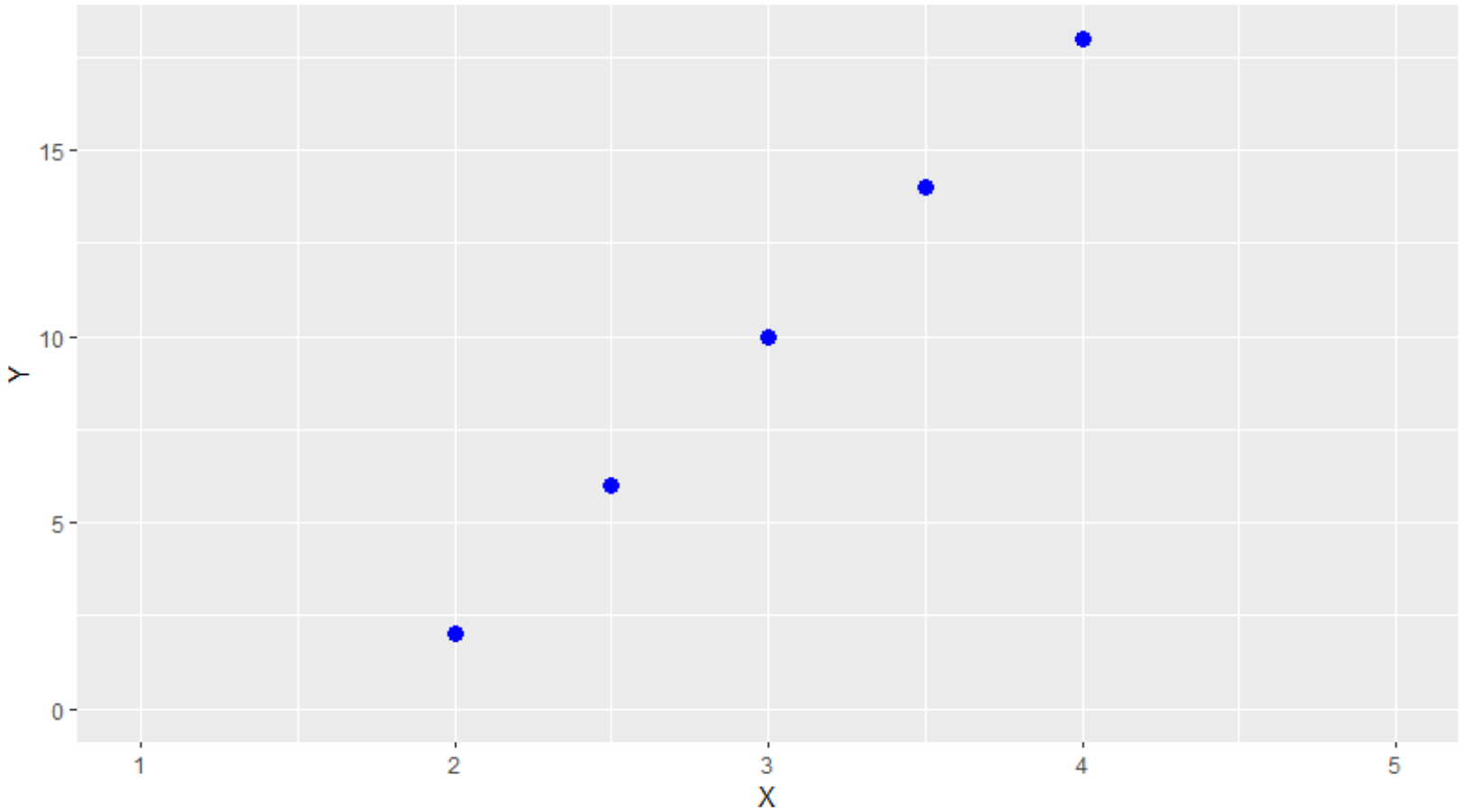

Final Thoughts

- Correlation Coefficient is a measure of *linear* association between two continuous variables
- Correlation Coefficient *value* represents *tightness* of linear relationship, not magnitude
- Correlation Coefficient is a building block for linear regression...

R Code

Correlation Coefficient

- cor(data$X, data$Y, use="pairwise.complete.obs")

Correlation Coefficient with Hypothesis Test

- cor.test(data$X, data$Y, method="pearson", conf.level=0.95)

Correlation Matrix (Matrix of Correlation Coefficients)

- cor(cbind(data$X, data$Y, data$Z, data$M, data$Q), use="pairwise.complete.obs")